

Spectral Signatures of Transformer Redundancy: From Attention-Head Pruning to Embedding Compression

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Abstract—Can inexpensive spectral statistics—computed directly from a transformer’s attention maps and embedding parameters with a single spectral transform—serve as a *general diagnostic for redundancy*, across both the attention maps a model computes internally and the embedding tables it stores? We test this diagnostic hypothesis on head pruning and embedding compression, and report where it holds, where it fails, and what that boundary teaches. For pruning, per-head subband statistics rank heads far better than popular mass heuristics: ranking by attention-map Frobenius norm or max-column mass is *anti-predictive* of ablation damage at single-head granularity. A leave-one-cell-out ridge model over these features is the strongest representation-level predictor we test, keeps its accuracy when trained *entirely on other architectures* (leave-one-model-out), and a 45-cell reference sweep confirms the ordering (87/90 wins vs. non-wavelet baselines, all significant under FDR control). Yet even this most rigorously validated predictor fails the test that matters: validated downstream, rankings reorder—on SST-2 the architecture-robust ridge degrades to composite-level damage while attention-distribution entropy stays safe; on GPT-2 the asymmetry inverts outright (a different mechanism than the tied-embedding failure we find on the compression side). Representation-level screening, however rigorous, does not certify task-level safety. For compression, an equal-effective-byte shootout shows int8 quantisation stacks multiplicatively with frequency-domain sparsification, product quantization dominates extreme budgets, PCA truncation never wins the representation-level comparison (though no method is uniformly safe downstream) — and DCT/FFT controls reveal the advantage is generic spectral sparsity rather than anything wavelet-specific. A downstream transfer study of compressed tables mirrors the pruning lesson: low degradation on the tested encoder-classification tasks for every sparse-int8 recipe, divergent — even catastrophic — under transform-domain coding on the tied-embedding GPT-2 decoder. Code, evaluation sets and offline regression tests will be released.

Index Terms—transformers, attention heads, pruning, spectral analysis, wavelet transform, model compression, product quantization

I. INTRODUCTION

Multi-head self-attention [1] accounts for a substantial share of the parameters and inference cost of modern transformers, and a large body of evidence shows that many heads are redundant [5], [6], [12]. Two practical questions follow. (i) *Which* heads can be removed safely? Existing answers either require gradients, behavioural probing, or heuristics—most commonly the total attention mass a head commands—whose reliability

we show is worse than chance exactly where it matters, in the low-pruning-fraction regime. (ii) Once a model is fixed, its token-embedding tables remain a large dense artifact (e.g., $30,522 \times 768$ float32 ≈ 90 MB for BERT); how should they be stored under tight byte budgets?

This paper asks whether the hypothesis survives contact with practice, and states it as a falsifiable diagnostic claim: *inexpensive spectral statistics of a representation predict how much redundancy that representation carries — and hence what can be removed or compressed — uniformly across representation types*. We probe the claim at two extremes of the redundancy spectrum. Attention maps whose removal is harmless should look spectrally redundant (concentrated subbands, low coefficient entropy); token embeddings that survive aggressive lossy coding should concentrate their information in few spectral components. The discrete wavelet transform (DWT) [19], [20] supplies the instrument — one forward pass supplies the attention maps, the embedding table is read directly, and a linear-time transform yields signatures that are cheap and training-free — and DCT/FFT controls later test whether *wavelets* specifically matter or whether any frequency-domain basis carries the signal (§V-H). Equally important, we map where the diagnostic breaks: its rankings are validated at the representation level, and we show by direct experiment that such validation does not automatically transfer to task-level damage for either pruning (§V-G) or table compression (§V-H).

Contributions.

- 1) Per-head wavelet signatures predict single-head ablation damage significantly better than mass heuristics; Frobenius-mass and max-column-mass rankings are sign-inverted (anti-predictive) at this granularity (§V-D).
- 2) A seed-replicated local benchmark (nine cells, twelve predictors including zero-shot proxies of published criteria and a leave-one-cell-out ridge model) plus a 45-cell reference sweep across five model families confirm the representation-level ordering at scale: 87/90 paired wins over non-wavelet baselines, all significant (§V-E, §V-F), and the ridge predictors keep their accuracy when trained leave-one-model-out (§V-E).
- 3) A downstream-task validation (fine-tuned SST-2 accuracy, GPT-2 perplexity) quantifies the representation-

to-task gap for the whole predictor family: true attention entropy transfers best to SST-2; the hand-weighted wavelet composite collapses; Frobenius-mass rankings invert between levels; and even the leave-one-model-out ridge — the strongest, most rigorously validated representation-level predictor — inherits the gap, showing that learning against representation damage does not substitute for task-level validation (§V-G).

- 4) A controlled compression shootout at equal *effective* bytes shows int8 quantisation stacks multiplicatively with sparsification, dense product quantization dominates below $\sim 10\%$ budgets, and PCA truncation never wins the representation-level comparison; DCT/FFT controls show the advantage is generic spectral sparsity, and the residual basis ranking follows table heavy-tailedness (§V-H, §V-I). A downstream transfer study of compressed tables mirrors the pruning lesson and exposes a tied-embedding failure mode (§V-H).

II. BACKGROUND AND RELATED WORK

A. Structured attention-head pruning

Michel et al. [5] showed that many heads can be removed after training with little effect and that importance varies by layer; Voita et al. [6] prune with a differentiable regulariser; Behnke and Heafield [7] combine lottery-style retraining with head dropping for machine translation; Lagunas et al. [8] extend block pruning to whole transformers; Sanh et al. [9] introduce movement pruning during fine-tuning; Molchanov et al. [10] popularised gradient/Taylor importance criteria. These approaches either modify training or require dedicated probes. Our predictors read only the attention maps of the frozen model, and we evaluate them precisely where practitioners apply them: ranking heads for zero-ablation, validated both representationally and on downstream tasks.

B. Redundancy and geometry of representations

Cordonnier et al. [11] observe that heads can converge to similar query/key roles; Dalvi et al. [12] quantify redundancy among neurons and layers; Ethayarajh [25] documents the anisotropic cone geometry of contextual embeddings. Our static-table measurements add a spectral view: BERT-family vocabularies are strongly heavy-tailed (excess kurtosis 12.9–22.5), whereas GPT-2’s near-isotropic table (kurtosis 0.13) behaves differently under every compression scheme.

C. Compression and quantization

Product quantization (PQ) [13] remains central to billion-scale nearest-neighbour search [14], with anisotropic refinements [15]; integer quantisation underpins deployed inference [16], [17]; low-rank adaptation exploits spectral structure for parameter-efficient tuning [18]. Wavelets have long served signal compression [19], [20], but have not been benchmarked head-to-head against PCA truncation, magnitude sparsification and PQ *at equal effective byte budgets* on transformer embedding tables.

D. Spectral analysis of learned representations

SVCCA [26] compares representations via singular vectors; spectral bias studies [27] show networks learn low frequencies first. Our results extend this line to *redundancy prediction*: subband statistics forecast what can be removed.

III. METHOD

A. Wavelet decomposition of embeddings and attention

Let $x \in \mathbb{R}^D$ be a token embedding ($D=768$, or 2048 for TinyLlama). We symmetrically pad x to the next power of two n , apply a level- L DWT with filter ψ (Haar, DB4, Sym4, Coif2; default DB4), obtaining approximation coefficients c_A and detail coefficients $c_{D,\ell}$, $\ell = 1..L$ (level 1 = finest), with subband energies $E_\ell = \|c_{D,\ell}\|^2$. Attention matrices receive the same machinery in 2-D (separable DWT2) on the row-normalised matrix, yielding (c_{AH}, c_{AV}, c_{AD}) subbands per level. Per head we record: total/approx/ low/high-band energies and their ratio (low = coarsest half of levels), Shannon entropy of normalised coefficient magnitudes, spectral entropy over level energies, Gini coefficient of magnitudes, reconstruction L2 error after zeroing the smallest 30% of detail coefficients, and the coefficient count retaining 99% energy.

B. Head-importance predictors

Each predictor maps per-head metrics to a score s_h (higher = predicted more important / less prunable): `wavelet_entropy` (s = coefficient entropy); a `wavelet` composite computed as the raw-scale sum $e + g + \frac{1}{2}r + \frac{1}{2}\rho$ over entropy e , Gini g , reconstruction error r and energy ratio ρ . We emphasise that this composite mixes unnormalised quantities (entropy spans several bits, r lives near 10^{-2}), so its effective weighting is dominated by e ; we retain it as the historically specified heuristic and *treat the ridge models below as its principled counterpart*; `frobenius_mass` ($\|A\|_F$ of the attention map); `max_column_mass` (mean over rows of each column’s maximum); `random`; `attention_entropy_true`, the Shannon entropy $-\sum_q \sum_c p_{qc} \log p_{qc}$ of the attention distribution itself. For controlled zero-shot comparison we additionally construct three attention-only *proxies inspired by published criteria*. These are deliberately *not* reproductions of the original task-aware, training-time algorithms; they exist so that every method sees exactly the same inputs under identical conditions: `micel_inspired` follows Michel et al.’s discardability notion [5] via the magnitude-peakedness proxy $\|A\|_F \cdot (1 - H(A)/\log T)$ — heads with near-uniform attention score as discardable; `voita_inspired` follows Voita et al.’s importance notion [6] as $\max_c \bar{A}_{\cdot c} \cdot (1 - r)$, i.e. kept attention mass discounted by the 30%-threshold wavelet reconstruction error r ; and a `behavioural_similarity` proxy scores each head by behavioural dissimilarity, $1 - \max_{h' \neq h, \text{layer}(h')=\text{layer}(h)} \cos(\text{vec}(A_h), \text{vec}(A_{h'}))$, so redundant near-copies rank as discardable. All three consume exactly the same per-head attention maps and metric rows as our predictors. Three ridge regressors fit on (a) wavelet features only, (b) distributional features only, (c) both, are trained *leave-one-cell-out* (LOOO): weights for every held-out cell are

fitted on all other cells’ (features, measured loss) pairs; inputs are standardised within each cell; $\alpha=1$. The ridge models answer the composite’s weighting problem by learning it.

C. Evaluation protocol and sign convention

Representation-level ground truth for one input is the per-head cosine drop between original and ablated sentence representations when head h alone is zero-ablated (hooks on the post-attention projection make this architecture-agnostic). Stored correlations use $\text{corr}(-s, \Delta)$ where Δ is the damage; because batched pruning removes the *lowest-scored* fraction, a good predictor has *negative* stored correlations. Orientation-free batched sweeps (damage vs. pruned fraction) complement the correlations. *Downstream ground truth*: SST-2 validation accuracy for fine-tuned BERT-base/DistilBERT classifiers and WikiText-2 perplexity for GPT-2, each re-measured with the bottom- $p\%$ heads (by each predictor’s ranking, computed from sample texts through the same frozen model) simultaneously ablated.

D. Compression schemes at equal effective bytes

All methods are compared at equal *total* storage per token: payload *plus* metadata *plus* amortised shared parameters. With vocabulary size V : PCA/random-projection store d floats ($4d$ B) plus their projection matrix (Dd $4/V$; PCA adds a mean row, D $4/V$); sparse variants store kept coefficients as value + uint16 index (6 B fp32, 3 B int8 + 2 B per-vector scale); PQ stores an m -byte code plus the codebook ($256 \cdot D \cdot 4/V$; 26 B/token for $D=768$, 65 B/token for $D=2048$ at $V \approx 30\text{k} - 50\text{k}$). At million-vector deployment scales this amortised overhead falls to roughly 0.8–2.1 B/token for $D=768-2048$, and continues to decrease with scale. Quality metrics: cosine between original and reconstructed vectors and Jaccard overlap of top-10 cosine neighbours before vs. after compression, over 1,500 held-out tokens per model (PCA/PQ fitted on a disjoint split).

IV. EXPERIMENTAL SETUP

Models. bert-base-uncased, distilbert-base-uncased (encoders; 144/72 heads), GPT-2 (decoder; 144 heads); generality sweep adds RoBERTa-base [21], DeBERTa-base [22] and TinyLlama-1.1B [23] ($D=2048$). **Data.** Static tables: full HF vocabularies. Contextual probe: a 10-anchor polysemy set. Attention/pruning: a curated 12-sentence eval set spanning polysemy, relative clauses, passives, named entities, numerals, questions; benchmark corpus: WikiText-2 sentences [24] filtered to clean single sentences ($\approx 70\text{k}$ pool). **Downstream.** textattack/bert-base-uncased-SST-2 and distilbert-base-uncased-finetuned-sst-2-english on GLUE SST-2 validation (872 examples); GPT-2 perplexity on the WikiText-2 test split. **Hardware.** All experiments run CPU-only (Intel i7-1355U); the pipeline supports containerised GPU sweeps. **Statistics.** Paired Wilcoxon tests, Cohen’s d , Cliff’s δ , bootstrap CIs, three seeds per cell. Where multiple paired comparisons are reported together we additionally apply Benjamini–Hochberg false-discovery-rate control at $\alpha=0.05$.

Unit of analysis. Every paired test operates on *per-input correlation values* — one Pearson r per sentence input, computed over all heads of that input — so the sample entering a Wilcoxon test is the set of inputs ($n=12$ per local cell, 150 per reference-sweep cell group), never individual heads (heads within one network are strongly dependent). The three seeds contribute disjoint sentence subsets drawn without replacement from the cleaned corpus, so pooling seeds widens coverage without duplicating inputs; seed-level variance is reported separately.

V. RESULTS

A. Static spectra differentiate model families

BERT/DistilBERT vocabularies are strongly heavy-tailed (excess kurtosis 12.9/22.5, skew $-1.95/-2.97$) while GPT-2’s are near-Gaussian (kurtosis 0.13); mean embedding energy differs by $8\times$ (21.1 vs. 2.6). Energy concentrates in the finest bands (low/high ratios 0.15–0.55): token identity lives in high-frequency detail. Zeroing the smallest 10%/50% of detail coefficients reconstructs at cosine $\geq 0.999/\geq 0.964$.

B. Spectral signatures track word sense

On the polysemy probe (DistilBERT, DB4), cross-sense spectrum separation tracks raw-cosine separation (bank 0.234 vs. 0.222) and same-sense similarity exceeds cross-sense similarity for all ten anchors (inter 0.56–0.84 vs. cross 0.29–0.68): spectra are sense-sensitive, though not more discriminative than cosine here.

C. Layer progression

Across DistilBERT layers, the low/high-frequency energy ratio rises ($1.35 \rightarrow 2.03$), Gini grows ($0.57 \rightarrow 0.71$), entropy falls ($7.16 \rightarrow 6.60$ bits) and 30%-threshold reconstruction error halves ($0.059 \rightarrow 0.028$): later layers attend through fewer, lower-frequency, more compressible patterns.

D. Single-head prediction: mass heuristics are anti-predictive

Table I reports stored correlations (negative = better). The wavelet composite and true attention entropy form the best tier; *Frobenius mass* and *max-column mass* carry positive correlations: their lowest-scored heads are the *most* damaging to remove. The orientation-free bottom-10% sweep makes this concrete — damage under lowest-score-first pruning: 0.00056 (attn-entropy-true), 0.00072 (wavelet), 0.00077 (wavelet-entropy), 0.00256 (random), 0.00318 (Frobenius mass), 0.00481 (max-column mass). Ranking by $\|A\|_F$ — a common shortcut — is $4.4\times$ worse than random here.

In batched bottom- $p\%$ sweeps (Fig. 1) the wavelet composite is best-or-tied-best in 5/12 model $\times$ ratio cells and never worse than random; the attention-entropy family wins most other cells. GPT-2 is nearly insensitive to single-head ablation at this probe (drops $\sim 10^{-4}$), compressing all differences there.

TABLE I

SINGLE-HEAD PRUNING PREDICTION (STORED PEARSON $r = \text{corr}(-s, \Delta)$; MORE NEGATIVE = BETTER). DB4, 12-SENTENCE EVAL SET. THE GPT-2 COLUMN IS COMPUTED ON A NEAR-NULLED DAMAGE SIGNAL (SINGLE-HEAD DROPS $\sim 10^{-4}$) AND SHOULD BE READ AS A NOISE FLOOR RATHER THAN A RANKING.

Predictor	BERT-base	DistilBERT	GPT-2
Attn. entropy (true)	-0.32	-0.40	-0.14
Wavelet composite	-0.31	-0.33	-0.16
Wavelet entropy	-0.14	-0.02	-0.18
Random	-0.02	+0.30	-0.02
Frobenius mass $\ A\ _F$	+0.33	+0.39	+0.16
Max-column mass	+0.33	+0.36	+0.19

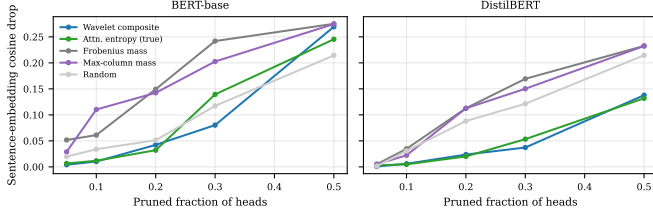


Fig. 1. Batched fairness sweep: mean sentence-representation cosine drop after ablating the bottom $p\%$ of heads ranked by each predictor (DB4, encoder models). Lower is safer.

E. Seed-replicated benchmark with learned combination

Across nine (model, dataset, seed) cells on WikiText-2 sentences (Table II), the *learned* predictors lead: the LOOO ridge over combined features is the best overall predictor on DistilBERT (-0.41), the wavelet-only ridge is best on BERT-base (-0.328), and the combined model statistically ties wavelet-only there ($+0.007$, $p = 0.30$) while lifting significantly over it on DistilBERT ($\Delta = -0.074$, $p = 5 \times 10^{-4}$). Against every non-learned comparator — mass heuristics and published-criterion-inspired proxies alike — the combined model wins with $p \leq 0.0024$ and $|d|$ up to 9 on both encoders. Wavelet-side rankings beat the comparators in 72/105 comparisons overall. Seed standard deviation is non-degenerate (0.003–0.048). We report the hand-weighted composite for continuity with the specification it came from, but treat the ridge models — which learn the weighting from data with no test-cell leakage — as the primary wavelet-feature predictors throughout.

Leave-one-model-out transfer. Because cells sharing a model may correlate, we re-fit the ridges under a harsher protocol: all weights are learned from the *other two architectures only* and applied to every input of the held-out model. Cross-architecture transfer holds: the combined ridge reaches mean stored $r = -0.28 / -0.40 / -0.23$ on held-out BERT/DistilBERT/GPT-2 (vs. $-0.32 / -0.41 / -0.15$ when its own architecture’s other seeds assist training), and wavelet-only features transfer at $-0.28 / -0.34 / -0.27$. The ordering of predictors is unchanged under this protocol; only GPT-2’s absolute values shift appreciably, consistent with its near-null damage signal.

TABLE II

LOCAL BENCHMARK MEAN STORED CORRELATIONS (MORE NEGATIVE = BETTER; WIKITEXT-2, SEEDS $\{0, 1, 2\}$, 12 INPUTS/CELL).

Predictor	BERT-base	DistilBERT	GPT-2
Ridge: combined	-0.320	-0.413	-0.153
Ridge: wavelet-only	-0.328	-0.338	-0.189
Ridge: attn.-only	-0.237	-0.327	-0.219
Attn. entropy (true)	-0.237	-0.327	-0.219
Wavelet entropy	-0.203	-0.125	-0.154
Wavelet composite (heuristic)	-0.117	-0.130	-0.213
Frobenius mass	+0.256	+0.332	+0.221
Max-column mass	+0.270	+0.332	+0.227
Random	+0.031	+0.278	-0.048

TABLE III

REFERENCE SWEEP (45 CELLS, GPU): POOLED MEAN STORED CORRELATIONS PER MODEL FAMILY (MORE NEGATIVE = BETTER).

Model	Wavelet	Coeff.-ent.	Frob.	ColMass	Rand.
BERT-base	-0.136	-0.238	+0.254	+0.262	-0.003
DistilBERT	-0.123	-0.172	+0.325	+0.321	+0.279
GPT-2	-0.224	-0.169	+0.226	+0.230	-0.050
RoBERTa	-0.138	-0.235	+0.155	+0.140	-0.039
TinyLlama	-0.216	-0.228	+0.178	+0.183	-0.006

F. Scale validation: 45-cell reference sweep

To address small-sample concerns we report a larger reference run executed on containerised GPUs: 45 cells = five model families (BERT-base, DistilBERT, GPT-2, RoBERTa-base, TinyLlama) $\times$ three corpora (WikiText-2, Penn Treebank, GLUE subset) $\times$ three seeds, with 150 inputs per cell group and the fixed non-ridge predictor set described in §III-B. The strength of this sweep lies less in any single p -value than in the convergence of evidence: five architectures, three corpora, disjoint per-seed inputs, and rankings that agree with the small-scale study. Table III pools dataset-level means per model. In the 90 paired comparisons between the two wavelet-side scores (composite, coefficient entropy) and the five non-wavelet baselines (Frobenius mass, max-column mass, random, Michel/Voita-inspired proxies), the wavelet side is better in **87/90** and *all 90* are statistically significant (Wilcoxon; every one survives Benjamini–Hochberg FDR control at $\alpha=0.05$, largest raw $p = 10^{-5}$). Across all 105 pairwise comparisons recorded by the harness, the wavelet side wins 91, with 103 significant (103/105 under BH-FDR). Seed standard deviations span 0.0001–0.0124. Two honest caveats: DeBERTa cells were produced before the architecture-agnostic ablation hook covered disentangled attention and are excluded; on TinyLlama the behavioural-similarity proxy is anomalously strong (-0.32), the only case where a non-wavelet-side ranking leads.

G. Downstream task validation

Table IV re-ranks the same predictors by *task-level* damage, and the picture inverts across objectives. On SST-2, true attention entropy is the most robust ranking (bottom-30% pruning costs 11.2/5.9 accuracy points on BERT/DistilBERT), while

the wavelet *composite* — strong at representation level — collapses (+41.5/+42.0 points). On GPT-2 perplexity the ordering flips entirely: the wavelet composite is by far the safest ranking (perplexity $1.25\times/2.7\times/3.3\times$ clean at 10/20/30%, vs. random at $12.8\times/62\times/39.5\times$), whereas attention entropy becomes catastrophic ($1.9\times/32.7\times/160\times$). Frobenius mass is likewise level-inconsistent: anti-predictive for representation damage yet among the best SST-2 rankings on DistilBERT (+0.6/+1.3/+4.9 points). Most tellingly, the entire *learned* ridge family — our strongest representation-level predictors under both LOOO and leave-one-model-out training — degrades downstream: the combined ridge loses 14.3/41.2/41.5 accuracy points on BERT, and even its attn-only variant, the gentlest learned ranking, tracks attention entropy rather than beating it. Learning the feature weighting against representation damage does not repair objective mismatch: an estimator optimised for that damage inherits its blind spots. The GPT-2 column adds an instructive asymmetry. The plain wavelet composite is the safest ranking there ($1.3\times/2.7\times/3.3\times$), yet the ridge fitted *on wavelet features alone* is far worse ($18\times/12\times/29\times$): fitting weights to representation-level loss destroyed whatever accidental alignment made the composite robust. Conversely the attn-only ridge hedges attention entropy’s blow-up ($2.6\times/3.0\times/5.8\times$ vs. $160\times$ at 30%) by diluting it with other features. Mixing hedges; it does not compose. Two conclusions follow. First, representation-level cosine damage is an imperfect proxy: predictor rankings reorder across downstream objectives and architectures — even when the predictor is learned against that proxy under rigorous protocols — so any pruning claim must include a task-level check matched to the deployment objective. Second, the right target for learned rankers is the damage metric a practitioner actually cares about: fit ridges on task-level losses directly.

H. Compression at equal effective bytes

Fig. 2 and Table VI compare methods at equal *total* storage. Three effects survive the stricter accounting. (i) *int8 stacks*: at matched bytes the int8 twin of a sparsifier keeps $2\times$ more coefficients than its fp32 counterpart for identical bytes and dominates everywhere (e.g., 1024 B: 0.95 vs. 0.83 cosine for raw top-k). (ii) *PQ owns the extreme regime*: at 128 effective bytes/token (24 $\times$ smaller than the original payload) it reconstructs at cosine 0.88 retaining 41% of exact top-10 neighbours; no per-vector method exceeds 0.54 there. Its codebook amortisation (26–65 B/ token at these vocabulary sizes) is included; at million-vector deployment scales the overhead falls to roughly 0.8–2.1 B/token for $D=768$ –2048, and its advantage widens further. (iii) *PCA never wins*: global principal components align poorly with per-token information; even untrained random projection is often competitive, and both trail sparsification except at saturating budgets. (iv) *The sparse-coding advantage is largely generic spectral sparsity, not wavelet-specific*: adding DCT-II (global frequency basis, no multiresolution) and rFFT controls at identical coefficient counts and byte charges, the DCT tracks or exceeds the DB4 wavelet column on cosine at every effective budget

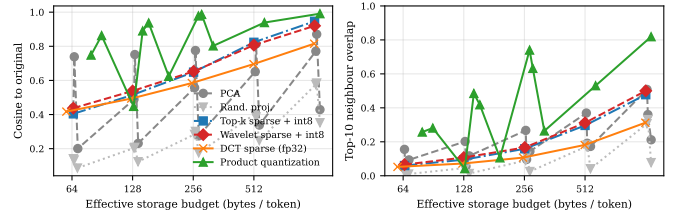


Fig. 2. Compression quality vs. effective storage budget (bytes/token including metadata and amortised shared parameters; original = 3072 B for $D=768$). Pooled over six models.

(pooled over six models, fp32 payloads: 0.50/0.59/0.70 vs. wavelet 0.44/0.58/0.69 at 128/256/512 B), with neighbour overlap statistically indistinguishable, and the complex-valued FFT trailing both only because its payload costs more per kept coefficient. Within transform-domain sparsification, the choice of orthonormal frequency basis contributes little; multiresolution itself provides no measurable advantage for fixed-length embedding vectors. The wavelet-vs-raw-topk question is decided by table geometry (§V-I), and the DCT/FFT controls bound how much of that is *wavelet-specific*: little.

From representation metrics to tasks. Applying the same lesson we learned for pruning, we replace each model’s word-embedding table with its lossy reconstruction and re-measure the task metric (Table V). On the fine-tuned SST-2 classifiers, *every* tested int8 sparsification recipe — wavelet, DCT, and raw coordinate top-k alike — transfers with low degradation (all drops ≤ 4.9 points at 17% and ≤ 0.7 points at 33%), confirming that the strong representation-level performance of sparse int8 coding transfers to these encoder tasks *without a unique advantage for the spectral bases*. At 33% storage every tested sparse-int8 method stays within 0.7 accuracy points on both encoders; at 17%, degradation remains modest but reaches 4.9 points. PCA truncation at matched effective bytes is dramatically worse (−36/−17 points at 33%) despite respectable cosine — exactly the representation-vs-task gap the pruning study exposed, now on the compression side. GPT-2 is harsher still, and here the bases separate: wavelet and DCT tables diverge outright ($> 10^3\times$ clean perplexity), PCA leaves 22–47 $\times$ degradation, while *raw-coordinate* top-k int8 at the 33% budget survives at $1.21\times$. We hypothesise two mechanisms: GPT-2’s *tied* input/output embedding matrix (*wte* doubles as the output softmax weight, so table noise flows straight into every logit) amplifies any transform that smears perturbations across all dimensions, whereas coordinate-space selection leaves each surviving embedding row individually intact. Untying/re-calibration experiments are left to future work. The practical consequence, stated with its scope: in the two encoder-classification settings tested, every tested int8 sparsification recipe transfers with little task degradation regardless of basis, while tied decoder tables require basis-aware choices and separate task-level validation — representation-level metrics alone would have hidden every one of these failures.

TABLE IV

DOWNSTREAM-TASK VALIDATION: DAMAGE AFTER PRUNING THE BOTTOM- $p\%$ OF HEADS RANKED BY EACH PREDICTOR. ACCURACY DROPS (POINTS) AND PERPLEXITY RATIOS ($\times$ CLEAN) COMPUTED AGAINST EACH MODEL’S UNPRUNED SCORE; LOWER IS BETTER. NO PREDICTOR DOMINATES: DISTRIBUTIONAL ENTROPY TRANSFERS BEST TO SST-2 BUT COLLAPSES ON GPT-2, WHERE THE WAVELET COMPOSITE EXCELS — TASK-LEVEL VALIDATION IS INDISPENSABLE.

Predictor	BERT SST-2 Δ acc.			DistilBERT SST-2 Δ acc.			GPT-2 PPL ratio		
	10%	20%	30%	10%	20%	30%	10%	20%	30%
Wavelet composite	+8.9	+35.6	+41.5	+6.5	+14.4	+42.0	1.25	2.66	3.26
Attn. entropy (true)	+1.2	+6.3	+11.2	+1.4	+5.4	+5.8	1.91	32.65	159.53
Frobenius mass $\ A\ _F$	+6.4	+5.5	+16.2	+0.6	+1.3	+4.9	2.39	5.15	6.43
Max-column mass	+1.8	+14.1	+18.8	+0.9	+1.8	+10.8	2.23	5.05	8.40
Random	+4.8	+39.0	+41.5	+4.0	+15.0	+42.0	12.75	61.99	39.49
Ridge: wavelet-only (LOMO)	+15.7	+41.5	+41.5	+7.7	+42.0	+42.0	18.31	11.67	28.84
Ridge: attn.-only (LOMO)	+5.7	+10.6	+11.2	+0.1	+2.8	+3.7	2.57	2.99	5.84
Ridge: combined (LOMO)	+14.3	+41.2	+41.5	+6.5	+22.8	+42.0	7.81	16.25	23.70
Clean model	92.4	—	—	91.1	—	—	39.4	—	—

TABLE V

DOWNSTREAM VALIDATION OF EMBEDDING-TABLE COMPRESSION: THE WORD-EMBEDDING MATRIX IS REPLACED BY ITS LOSSY RECONSTRUCTION (INT8 PAYLOADS; PCA DENSE) AND THE TASK METRIC RE-MEASURED. ACCURACY DROPS (POINTS) AND PERPLEXITY RATIOS ($\times$ CLEAN); LOWER IS BETTER. ON BOTH ENCODER CLASSIFIERS EVERY TESTED INT8 SPARSIFICATION RECIPE — WAVELET, DCT, AND RAW TOP-K ALIKE — TRANSFERS WITH LOW DEGRADATION, AND IS NEAR-LOSSLESS AT THE 33% BUDGET — WHILE PCA TRUNCATION DOES NOT. ON TIED-EMBEDDING GPT-2 THE TRANSFORM-DOMAIN TABLES FAIL CATASTROPHICALLY AT BOTH BUDGETS WHILE RAW-COORDINATE TOPK STAYS ROBUST AT 33% ($1.2\times$); WHERE A PERTURBATION LANDS CAN MATTER MORE THAN HOW CLOSE THE RECONSTRUCTION IS.

Model	Wav.+int8		DCT+int8		TopK+int8		PCA	
	17%	33%	17%	33%	17%	33%	17%	33%
BERT SST-2	+1.9	+0.3	+0.7	+0.3	+0.8	+0.6	+39.0	+36.0
DistilBERT SST-2	+4.9	+0.1	+1.1	+0.7	+1.5	+0.5	+25.2	+17.1
GPT-2 PPL	$>1000\times$	$>1000\times$	$>1000\times$	$>1000\times$	$>1000\times$	$1.2\times$	$46.7\times$	$21.6\times$

TABLE VI

EQUAL-*effective*-BYTE COMPARISON POOLED OVER SIX MODELS (COSINE / TOP-10 NEIGHBOUR JACCARD). BEST SPARSE METHOD PER ROW BOLDDED.

B_{eff}	PCA	Rand.proj.	TopK-int8	Wav-int8	PQ
128	.53/.14	.19/.04	.52/.09	.54/. 11	.88/.41
192	.56/.17	.24/.07	.59/.13	.61/. 14	.89/.48
256	.58/.19	.27/.09	.65/.16	.65/. 17	.93/.59
384	.69/.25	.35/.14	.76/. 25	.76/. 25	—
512	.72/.30	.39/.17	.82/.30	.81/. 31	—
1024	.81/.44	.58/.31	.95/.48	.92/. 50	—

I. Generalisation to other architectures

Repeating the shootout on RoBERTa-base, DeBERTa-base and TinyLlama (2048-dim decoder; Table VII), two orderings generalise universally and one is conditional: int8 stacking strictly dominates its fp32 twin on every model at every budget; PQ dominates the extreme regime on every model; PCA never wins overall. Conditionally, spectral sparsification (wavelet or DCT alike) beats naive coordinate-space magnitude selection most clearly on the heavy-tailed BERT-family tables, while on isotropic tables (RoBERTa, DeBERTa, TinyLlama) raw top-k with int8 payloads becomes the stronger sparse method, and the residual wavelet-vs-DCT gap is within noise. Heavy-tailedness is measurable in one pass over a sample, giving practitioners a decision rule for basis choice — though given §V-H, the safer summary is that basis choice is secondary to selecting a sparsification strategy matched to

TABLE VII

GENERALISATION OF THE COMPRESSION RANKING TO UNSEEN ARCHITECTURES AT EQUAL *effective* STORAGE BUDGETS (COSINE / TOP-10 NEIGHBOUR OVERLAP; BYTES/TOKEN INCLUDING ALL METADATA AND AMORTISED SHARED PARAMETERS). BEST SPARSE METHOD PER ROW BOLDDED; — = METHOD CANNOT FIT THE BUDGET.

Model	B_{eff}	PCA	TopK-int8	Wav-int8	PQ
RoBERTa-base	128	0.42/0.23	0.55/0.13	0.48/0.07	0.85/0.44
	256	0.50/0.29	0.69/0.26	0.62/0.16	0.97/0.72
	512	0.62/0.39	0.85/0.45	0.77/0.31	—
DeBERTa-base	128	0.44/0.22	0.54/0.10	0.49/0.07	0.85/0.43
	256	0.52/0.28	0.68/0.19	0.63/0.13	0.97/0.71
	512	0.62/0.37	0.83/0.37	0.77/0.29	—
TinyLlama-2048	128	0.23/0.11	0.39/0.04	0.36/0.03	—
	256	0.27/0.14	0.50/0.07	0.48/0.05	0.72/0.19
	512	0.33/0.17	0.63/0.12	0.62/0.10	0.91/0.48

the geometry and downstream use of the embedding table. In Table VII, “—” marks budgets a method cannot meet (for PQ on TinyLlama, the code plus the amortised codebook exceeds the 128 B/token budget for every evaluated configuration) or points outside the evaluated range.

VI. LIMITATIONS

Our pruning ground truth is representational (sentence-embedding cosine drop) *and* behavioural (SST-2 accuracy, language-model perplexity), but the behavioural set covers one classification task and one LM evaluation; broader task suites

may reorder predictors again. The 45-cell reference sweep used the fixed non-ridge predictor set (no LOOO ridge) and excludes DeBERTa (its cells predate the architecture-agnostic ablation hook); DeBERTa appears only in the compression sweep. Local benchmark cells contain twelve inputs each; we mitigate with seeds, paired tests and the large reference sweep, but absolute correlation magnitudes at this scale should be read cautiously. GPT-2’s insensitivity to single-head ablation bounds any predictor’s measurable advantage there. The leave-one-model-out ridge study covers three architectures; extending it to the five-family scale is future work. Finally, PQ requires training data and fixed codebooks; for streaming single-vector storage, sparse int8 coding remains the recommended operating regime, with the sparsification basis selected according to table geometry and downstream use — subject to the tied-embedding caveat of §V-H.

VII. CONCLUSION

We asked whether inexpensive spectral statistics act as a general redundancy diagnostic across transformer representations. The answer is a qualified yes with a sharp boundary. As a *diagnostic*, spectral signatures rank attention heads for pruning far better than mass heuristics (which are anti-predictive at single-head granularity), transfer across architectures under leave-one-model-out training, and hold at 45-cell scale; on the compression side they characterise which sparsification formats perform best at the representation level across effective budgets, with DCT/FFT controls showing the effect is generic spectral sparsity rather than something unique to wavelets. As a *deployment criterion*, the diagnostic is insufficient — by design it measures representation damage, and our downstream studies show rankings reorder between that objective and task accuracy or perplexity, with catastrophic failures observed on the tied-embedding GPT-2 decoder — and they do so even when the ranking is *learned* against representation-level damage rather than hand-tuned. Spectral pre-screening plus task-level validation is therefore the recommended workflow. Code, evaluation sets and offline regression tests will be released upon publication.

REFERENCES

- [1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems*, 2017, pp. 5998–6008.
- [2] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.
- [3] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, “Language models are unsupervised multitask learners,” OpenAI, Tech. Rep., 2019.
- [4] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, “DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter,” in *NeurIPS EMC² Workshop*, 2019.
- [5] P. Michel, O. Levy, and G. Neubig, “Are sixteen heads really better than one?” in *Advances in Neural Information Processing Systems*, 2019, pp. 14 014–14 024.
- [6] E. Voita, D. Talbot, F. Moiseev, R. Sennrich, and I. Titov, “Analyzing multi-head self-attention: Specialized heads do the heavy lifting, the rest can be pruned,” in *Proc. ACL*, 2019, pp. 5797–5808.

- [7] M. Behnke and M. Heafield, “Losing heads in the lottery: Pruning transformer attention in neural machine translation,” in *Proc. EMNLP-IJCNLP*, 2021, pp. 4942–4952.
- [8] F. Lagunas, E. Charlaix, V. Sanh, and A. M. Rush, “Block pruning for faster transformers,” in *Findings of EMNLP*, 2021, pp. 3578–3595.
- [9] V. Sanh, T. Wolf, and A. M. Rush, “Movement pruning: Adaptive sparsity by fine-tuning,” in *Advances in Neural Information Processing Systems*, 2020, pp. 20 378–20 389.
- [10] D. Molchanov, S. Tararev, and J. Kautz, “Pruning convolutional neural networks for efficient inference,” in *Proc. ICML*, 2017, pp. 2610–2619.
- [11] J.-B. Cordonnier, A. Loukas, and M. Jaggi, “Multi-head attention: Collaborate instead of concatenate,” arXiv:2006.16362, 2020.
- [12] F. Dalvi, N. Durrani, H. Sajjad, Y. Belinkov, A. Bau, and S. Vogel, “Analyzing redundancy in pretrained transformer models,” in *Proc. EMNLP*, 2020, pp. 4908–4915.
- [13] H. Jégou, M. Douze, and C. Schmid, “Product quantization for nearest neighbor search,” *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 33, no. 1, pp. 117–128, 2011.
- [14] J. Johnson, M. Douze, and H. Jégou, “Billion-scale similarity search with GPUs,” *IEEE Trans. Big Data*, vol. 7, no. 3, pp. 535–547, 2021.
- [15] R. Guo, P. Sun, E. Lindgren, Q. Yang, F. Bian, X. Li, and O. Gurev, “Accelerating large-scale inference with anisotropic vector quantization,” in *Proc. ICML*, 2020, pp. 3880–3889.
- [16] B. Jacob, S. Kligys, B. Chen, M. Zhu, M. Tang, A. Howard, H. Adam, and D. Kalenichenko, “Quantization and training of neural networks for efficient integer-arithmetic-only inference,” in *Proc. CVPR*, 2018, pp. 2704–2713.
- [17] T. Dettmers, M. Lewis, Y. Belkada, and L. Zettlemoyer, “LLM.int8(): 8-bit matrix multiplication for transformers at scale,” in *Advances in Neural Information Processing Systems*, 2022, pp. 30 311–30 324.
- [18] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-rank adaptation of large language models,” in *Proc. ICLR*, 2022.
- [19] S. G. Mallat, “A theory for multiresolution signal decomposition: The wavelet representation,” *IEEE Trans. Pattern Analysis and Machine Intelligence*, vol. 11, no. 7, pp. 674–693, 1989.
- [20] I. Daubechies, *Ten Lectures on Wavelets*. Philadelphia, PA, USA: SIAM, 1992.
- [21] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, “RoBERTa: A robustly optimized BERT pretraining approach,” arXiv:1907.11692, 2019.
- [22] P. He, X. Liu, J. Gao, and W. Chen, “DeBERTa: Decoding-enhanced BERT with disentangled attention,” in *Proc. ICLR*, 2021.
- [23] P. Zhang, G. Zeng, T. Li, and H. Wang, “TinyLlama: An open-source small language model,” arXiv:2401.02385, 2024.
- [24] S. Merity, C. Xiong, J. Bradbury, and R. Socher, “Pointer sentinel mixture models,” in *Proc. ICLR*, 2017.
- [25] K. Ethayarajh, “How contextual are contextualized word representations? Comparing the geometry of BERT, ELMo, and GPT-2 embeddings,” in *Proc. EMNLP-IJCNLP*, 2019, pp. 55–65.
- [26] M. Raghu, J. Gilmer, J. Yosinski, and J. Sohl-Dickstein, “SVCCA: Singular vector canonical correlation analysis for deep learning dynamics and interpretability,” in *Advances in Neural Information Processing Systems*, 2017, pp. 6076–6085.
- [27] N. Rahaman, A. Baratin, D. Arpit, F. Draxler, M. Lin, F. Hamprecht, Y. Bengio, and A. Courville, “On the spectral bias of neural networks,” in *Proc. ICML*, 2019, pp. 5301–5310.